A UNIFORM SQUARE-ROOT EDGE-DEFICIT BOUND FOR STRONGLY CRITICAL GRAPHS AVOIDING FOUR- AND EIGHT-CYCLES

ARGUS OPEN RESEARCH PROJECT

Abstract. Motivated by the Erdős–Gyárfás conjecture that every graph of minimum degree at least three contains a cycle of length a power of two, we prove a parameter-uniform quadratic order bound in terms of the edge deficit $d = 2n - m$. Let G be a nonempty finite simple graph of minimum degree at least three containing no simple C_4 or C_8 such that no proper subgraph of G (including non-induced subgraphs) has minimum degree at least three. By combining component-sponsorship budgeting on vertices of degree at least five with an exact incidence identity that cancels all vertices of degree four, we prove that $n \le \max\{8d, 4d^2 - 8d\} \le 4d^2$. Consequently, any lexicographically minimal counterexample to the Erdős–Gyárfás conjecture satisfies $m \le 2n - \max\{7, \lceil 1 + \sqrt{n+4}/2 \rceil\}$.

1. Main Mathematical Results

Theorem 1.1 (Uniform square-root edge deficit)**.** *Let G be a nonempty finite simple undirected graph with $\delta(G) \ge 3$, no simple cycle of length 4 or 8, and no proper subgraph $H \subsetneq G$ with $\delta(H) \ge 3$. Put $n = |V(G)|$, $m = |E(G)|$, and $d = 2n - m$. Then $d \ge 2$ and*

$$n \;\le\; \max\{8d,\; 4d^2 - 8d\} \;\le\; 4d^2.$$

In particular, every minimum-order, minimum-size counterexample to the Erdős–Gyárfás conjecture satisfies $m \le 2n - \max\{7,\; \lceil 1 + \frac{\sqrt{n+4}}{2} \rceil\}$.

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1417 in the domain of extremal and structural graph theory adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Every nonempty finite simple graph G of minimum degree >= 3 with no proper subgraph of minimum degree >= 3 and no simple C_4 or C_8 satisfies n <= max(8d, 4d^2-8d) <= 4d^2 where d = 2n - m.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Unrestricted Erdos-Gyarfas power-of-two cycle conjecture for all minimum-degree-three graphs

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 1417 (extremal and structural graph theory), archived in the accompanying repository package `results/1417/`.

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